

Pratheek Hadse Pandesha

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EDUCATION

- 2021- 2026 **Washington University in St. Louis**
(expected) Ph.D. Candidate, **Donald Danforth Plant Science Center** |
Advisor: [R. Keith Slotkin](#) | Thesis: Primary small RNAs that initiate epigenetic silencing in Arabidopsis
- 2018-2020 **Indian Agricultural Research Institute, New Delhi**
M.Sc., in Plant Physiology | CGPA – 8.65 | [Gold Medal](#)
Advisor: [Viswanathan Chinnusamy](#) | Thesis: Investigating the role of DUF1645 proteins in abiotic stress tolerance and grain yield in rice
- 2014-2018 **University of Agricultural Sciences, Bangalore**
B.Sc., Agriculture Biotechnology | CGPA-8.66
Advisor: [R. Sowdhamini](#) | Thesis title: Structural and functional annotation of multiple stress responsive hypothetical proteins in rice

RESEARCH AND PROFESSIONAL EXPERIENCE

- 2021-2022 **Washington University in St. Louis / Donald Danforth Plant Science Center** (*Graduate research rotations*)
- Targeted methylation using CRISPR / SunTag system for viral disease resistance in Arabidopsis | Advisor: [Rebecca Bart](#)
 - Effect of nitrogen on circadian regulation of lipid metabolism in Arabidopsis | Advisor: [Dmitri Nusinow](#)
 - RNA-guided DNA insertion with CRISPR-associated transposases | Advisor: [R. Keith Slotkin](#)
- 2020-2021 **Bayer Crop Science**
Competitive Intelligence Analyst, Science Competitive Intelligence & Prospecting
- 2019-2020 **Massachusetts Institute of Technology (MIT), Cambridge, USA**
Khorana Scholar at the Dept. of Biology & Whitehead Institute for Biomedical Research
Advisor: [Mary Gehring](#) | Title: Pathways regulating genome dosage sensitivity in the endosperm of Arabidopsis
- 2017-2018 **National Centre for Biological Sciences (NCBS), Bangalore**
Bachelor's Dissertation
Advisor: [R. Sowdhamini](#) | Thesis title: Structural and functional annotation of multiple stress responsive hypothetical proteins in rice
- 2017 **Centre for Cellular and Molecular Biology (CCMB), Hyderabad, India**
JNCASR- Summer Research Fellow | Advisor: [Imran Siddiqi](#)
Development of fluorescent tag-based cell cycle marker system in Arabidopsis

- 2016 **University of Delhi, (South Campus), New Delhi, India**
Indian Academy of Sciences Summer Research Fellow | Advisor: Surekha Agarwal
 Identified and cloned a tetraspanin gene from *Oryza sativa*.
- 2016 **University of Agricultural Sciences, Bangalore, India**
Summer student | Advisor: Veena S. Anil, Dept. of Plant Biotechnology
 Standardized media composition and tissue culture protocol for somatic embryogenesis of Potato (Var. Kufri Jyothi- Late blight resistant cultivar)

PEER REVIEWED PUBLICATIONS

- Liu, P., Panda, K., Edwards, S.A., Swanson, R., Yi, H., **Pandesha, P.**, Hung, Y.H., Klaas, G., Ye, X., Collins, M.V. and Renken, K.N., 2024. Transposase-assisted target-site integration for efficient plant genome engineering. *Nature*, 631(8021), pp.593-600.
- Adavi B S, **Pandesha PH**, B J, Jha SK, Chinnusamy V, Sathee L. Nitrate supply regulates tissue calcium abundance and transcript level of Calcineurin B-like (CBL) gene family in wheat. *Plant Physiol Biochem*. 2023 Jun; 199:107724. doi: 10.1016/j.plaphy.2023.107724.
- Vineeth TV, Krishna GK, **Pandesha PH**, Sathee L, Thomas S, James D, Ravikiran KT, Taria S, John C, Vinaykumar NM, Lokeshkumar BM. Photosynthetic machinery under salinity stress: Trepidations and adaptive mechanisms. *Photosynthetica*. 2023 Apr 13;61(1):73-93.
- Sathee L, Jagadhesan B, **Pandesha PH**, Barman D, Adavi B S, Nagar S, Krishna GK, Tripathi S, Jha SK, Chinnusamy V. Genome Editing Targets for Improving Nutrient Use Efficiency and Nutrient Stress Adaptation. *Front Genet*. 2022 Jun 14; 13:900897.
- Pandesha PH**, Nagar S, Advani S, Chinnusamy V, Pandey R. Functional annotation of hypothetical proteins involved in multiple stress response in *Oryza sativa*. 2021; *Ind J Agr Sci* 91, 7.

MANUSCRIPTS UNDER REVIEW

- Montgomery, J.S., Soni, N., Marques Hill, S., Morran, S., Patterson, E.L., Edwards, S.A., Ratnayake, S., Hung, Y.H., **Pandesha, P.H.**, Slotkin, R.K. and Napier, R., 2024. A transposable element insertion in IAA16 disrupts splicing and causes dicamba resistance in *Bassia scoparia*. *bioRxiv*, pp.2024-07. (under review in *The Plant Journal*)
- Pandesha, P.H.**, Sajeevan, S., Sandeep, A., Mohamed. S. K., Pandey, R., Sathee, L., Ramanathan, Sowdhamini., Chinnusamy, Viswanathan. DUF1645 is a novel intrinsically disordered transcription factor family specific to plants. (under review in *New Phytologist*)

CONFERENCE PRESENTATIONS

- Pandesha, P.H.**, Edwards, S.A., Ratnayake, S., Slotkin RK. Quest for primary sRNAs that initiate epigenetic silencing in plants, *ICAR 2024, San Diego*; 15-19 July 2024; (Poster)
- Pandesh P.H.**, Sandip Adavi, Lekshmy Sathee, R. Sajeevan, Rakesh Pandey, Sowdhamini R, Viswanathan Chinnusamy. DUF1645- a novel transcription factor family in plants. *International Plant Physiology Congress (IPPC), Lucknow, India*; 2-5th December 2018; (Poster)

FELLOWSHIPS AND GRANTS

2022 - 2026	William H. Danforth Plant Sciences Fellowship, Washington University in St. Louis
2021	Trinity Henry-Barlow Fellowship and Cambridge Trust Scholarship for doctoral studies at the University of Cambridge, Cambridge Trust (Not availed)
2020 - 2021	ASPB Conviron Scholar, American Society of Plant Biologists
2018 - 2020	Junior Research Fellowship, Indian Council of Agricultural Research (ICAR)
2019	Khorana Scholar at MIT, funded by IUSSTF, United States Government and Department of Biotechnology, Government of India
2018	DBT- JNU Fellowship (for master's), Department of Biotechnology, Govt. of India (Not availed)
2014 - 2018	Merit Scholarship, University of Agricultural Sciences, Bangalore
2014 - 2018	Fee Concession and Merit Scholarship, Government of Karnataka, India
2014 - 2018	SDM Education Trust Scholarship, SDM Education Trust
2017	JNCASR- Summer Research Fellowship, Jawaharlal Nehru Center for Advanced Scientific Research
2016	Indian Academy of Sciences Summer Research Fellowship, Indian Academy of Sciences

AWARDS AND HONORS

2024	Best oral presentation award- WashU-Plant and Microbial Biosciences retreat
2024	Danforth Center CSTM professional development award
2024	The International Conference on Arabidopsis Research, San Diego- Registration Fee Award, The North American Arabidopsis Steering Committee (NAASC)
2021	Gold medal and best student of the year, Indian Agricultural Research Institute, New Delhi
2020	ASPB Plant Biology 2020 (Virtual) Registration Fee Award, American Society of Plant Biology
2019	Best poster award - National Agricultural Science Congress, Indian Agricultural Research Institute, New Delhi
2017	Travel grant- Nobel Prize Series, India by The Nobel Foundation, Sweden
2017	Travel grant- Indo-German Workshop on Plant-Insect Interactions Across Gradients, NCBS, Bangalore
2017	Travel grant-Advanced Biology Lecture Series (Theme: Gut Microbiome), NCBS, Bangalore
2016	Travel grant-Advanced Biology Lecture Series (Theme: Cancer Biology), NCBS, Bangalore
2014	Inspire Internship, Department of Science and Technology, Government of India

PROFESSIONAL ASSOCIATIONS AND SERVICES

2024 - Present	Member, North American Arabidopsis Steering Committee (NAASC)
2024 - Present	Arts & Sciences Graduate school cross program peer mentor, Washington University

2023 - Present	Co-chair, DBBS Student Advisory Committee, Washington University in St. Louis
2023 - Present	Member, National Center for Faculty Development & Diversity (through WashU-DBBS)
2022 - 2023	DBBS peer mentor, Washington University in St. Louis
2022	Graduate Teaching Assistant, BIO112-Introduction to Problem-Based Learning in Biology, Washington University in St. Louis
2020 - Present	Member, American Society of Plant Biologists
2020 - 2023	Scholar Mentor, WINStep Forward
2020	Volunteer, GCCR-Global Survey on Respiratory Illnesses and Changes in Smell, Taste, and Flavor in response to COVID-19, Bangalore
2018 - Present	<i>Co-founder, Next Generation Scientist Foundation (NGSF)</i> , India; Working towards providing internship opportunities at world-class labs for talented undergraduates from challenging social and economic backgrounds (www.ngsf.in)
2017	<i>Rural Agricultural Work Experience Program, Karnataka, India</i> Work focused on analyzing the effect of late blight of potato and soft rot of ginger on shifting cropping patterns in rural parts of Karnataka state in India.
2016 - 2017	Student ambassador, iBiology, Bangalore